

CANDIDATE NAME (CG) _____

INDEX NUMBER _____



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2011

BIOLOGY
Higher 2

Paper 2

Friday
12 August 2011

2 hours

Additional materials:
Answer paper

READ THESE INSTRUCTIONS FIRST

Write your name and index number in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in all the sections.

For Section A, write your answers in spaces provided on the question paper.

For Sections B, write your answers in the writing paper provided.

At the end of examination,

1. fasten all your work securely together;

FOR EXAMINER'S USE	
Section A	
1	/9
2	/9
3	/11
4	/9
5	/9
6	/12
7	/11
8	/10
Section B	
9/10	/20
TOTAL	/100

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 23 printed pages and 1 blank page

Section A (80 marks)

Answer all questions in the space provided

Question 1

Fig. 1.1 shows a cell undergoing two different stages of meiosis (to the same scale).

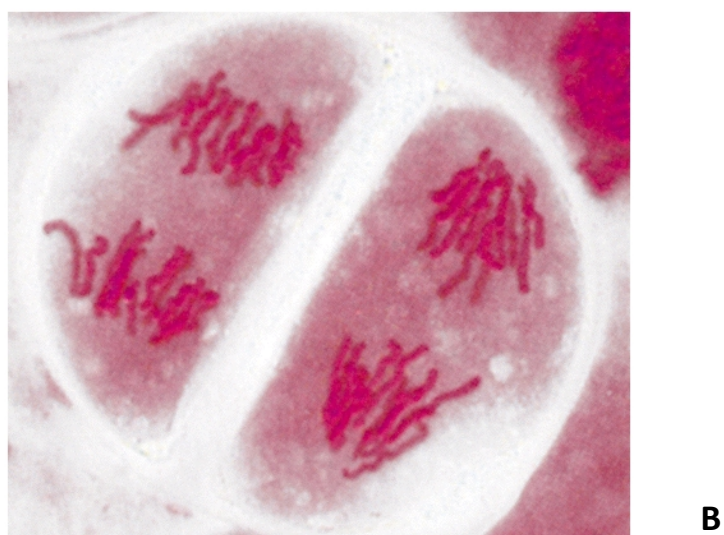
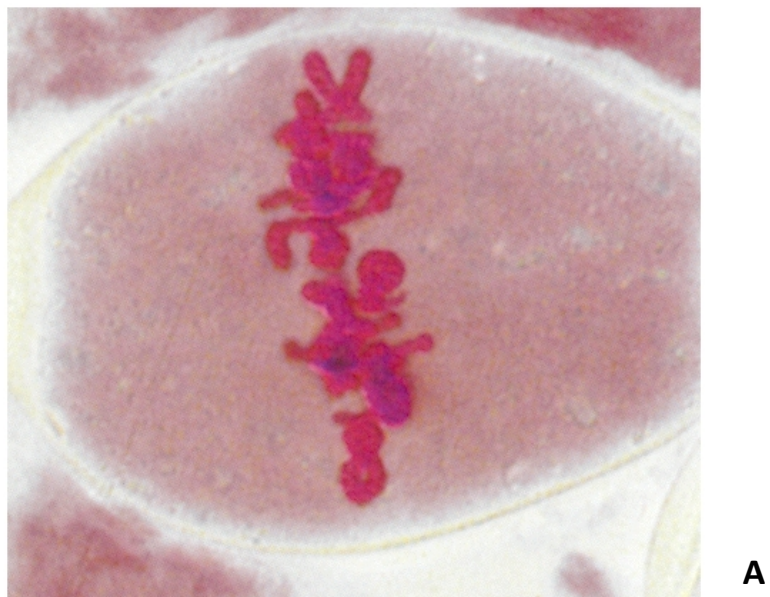


Fig. 1.1

(a) Identify stages **A** and **B**. [1]

Stage A

Stage B

- (b) Explain the phenomenon observed in **B**. [2]

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- (c) Describe how meiosis can lead to genetic variation. [2]

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Okadaic acid is a toxin that causes spindle fibres to significantly increase in length and disrupts the interactions between microtubules and kinetochores, resulting in a disorganized metaphase plate.

- (d) Suggest a possible outcome of a disorganised metaphase plate. [1]

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The chromosome carrying alleles **F/f** also carries alleles **E/e**, as shown in **Fig 1.2**.

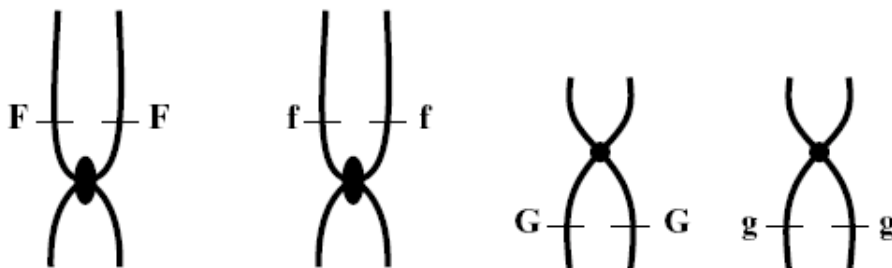


Fig. 1.2

- (e)(i) State the genotype of this cell. [1]

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- (ii) Explain how there are two copies of each allele of each gene at this stage of nuclear division. [2]

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[Total: 9 marks]

Question 2

Fig. 2.1 shows a molecule of tRNA and the enzyme that attaches the correct amino acid to it.

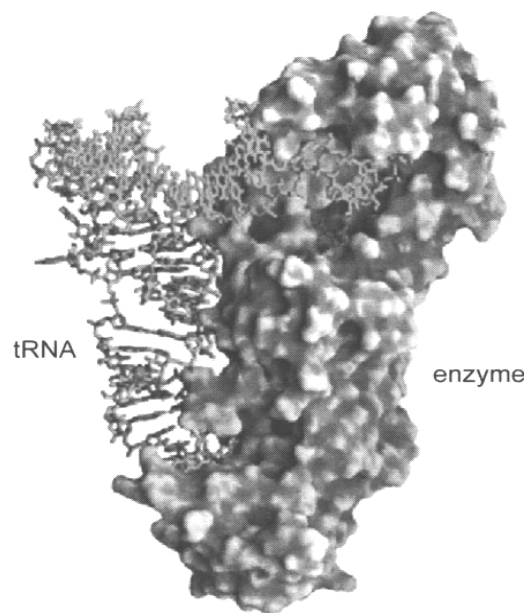


Fig. 2.1

(a) Describe **two** structural features which adapt tRNA to its role in translation. [2]

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(b) Explain how the enzyme in **Fig. 2.1** is suited to its function. [3]

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- (c) **Fig. 2.2** below shows the base sequence on a length of mRNA. The key indicates which amino acids they represent.

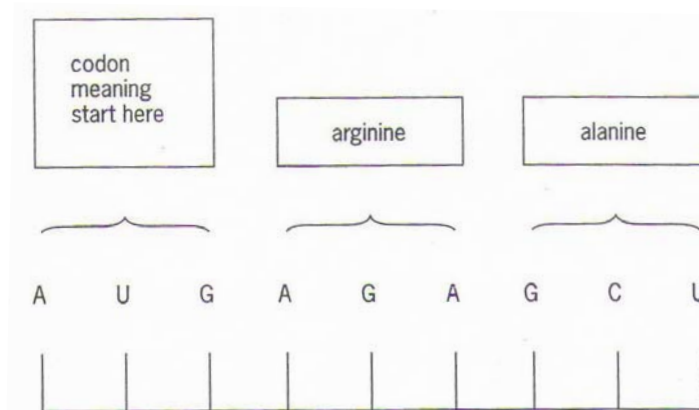


Fig. 2.2

The amino acid arginine is shown here as AGA but can also be represented by the codon AGG.

- (i) Explain why the substitution of A by G in this case does not affect the amino acid represented. [1]

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- (ii) The codons in the mRNA sequence do not overlap. Suggest 2 advantages of this. [2]

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Some genetic drugs are short sequences of nucleotides. They act by binding to selected sites on DNA or mRNA molecules and preventing the synthesis of disease-related proteins. There are two types:

1. triplex drugs are made from DNA nucleotides and bind to the DNA forming a three stranded helix.
2. antisense drugs are made from RNA nucleotides and bind to mRNA.

(iii) Name the process in protein synthesis which will be inhibited by [1]:

Drug type	Process inhibited
Triplex drugs	
Antisense drugs	

[Total: 9 marks]

Question 3

Fig. 3.1 shows the exchanges of substances between a mitochondrion and a chloroplast in a palisade mesophyll cell.

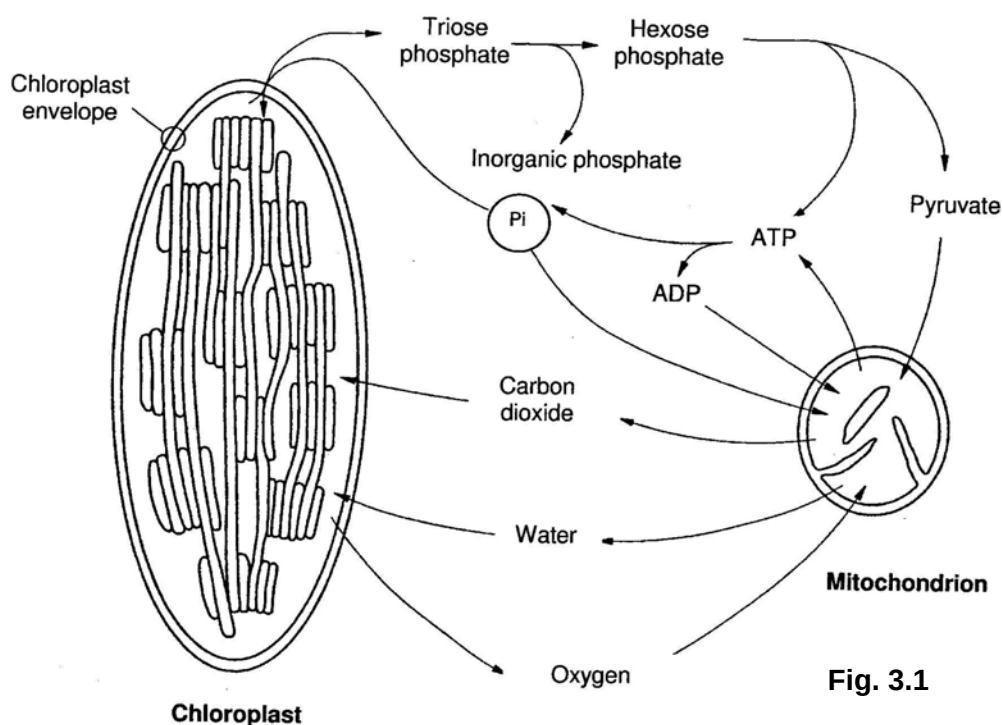


Fig. 3.1

A leafy shoot is sealed inside an air-tight transparent container. The concentration of oxygen in the atmosphere within this container can be measured. In the dark, the oxygen concentration falls. At high light intensities, the oxygen concentration increases. At a particular light intensity, the oxygen concentration in the container remains constant.

(a) With reference to **Fig. 3.1**, outline how the following links the metabolic pathways between the chloroplast and mitochondrion of this cell.

- (i) triose phosphate and hexose phosphate; [2]

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- (ii) gaseous exchange. [2]

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(b) With reference to **Fig. 3.1**,

(i) explain how it is possible for the oxygen concentration to remain constant. [1]

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(ii) chloroplasts do not have the transporters that allow them to export ATP to the cytosol. How then does the rest of the cell get the ATP it needs for cellular metabolism? [2]

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In plants, the carbohydrates manufactured are channeled to form two types of macromolecules, namely cellulose and starch.

(c) State the two structural differences between cellulose and starch. [2]

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(d) Explain how these differences allow them to serve different functions in a plant cell. [2]

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[Total: 11 marks]

Question 4

In an experiment, pea plants with round, black seeds were crossed with pure-breeding pea plants with wrinkled, green seeds. The offspring all had round, black seeds. These seeds were grown and the resultant plants allowed to self-pollinate (fertilise).

- (a)(i) What would be the expected phenotypic ratio? Using appropriate symbols draw a genetic diagram to explain your results. [5]

The above cross in (a)(i) produced 324 offspring with the following characteristics, as shown in **Table 4.1**.

Number of offspring	Phenotype
181	round, black seeds
51	round, green seeds
53	wrinkled, black seeds
39	wrinkled, green seeds

Table 4.1

- (ii) Complete the table below, **Table 4.2**, and calculate the chi-square to test the significance of the differences between the observed and expected results. [2]

	round, black seeds	round, green seeds	wrinkled, black seeds	wrinkled, green seeds
Observed number (O)	181	51	53	39
Expected number (E)				
Difference between observed & expected numbers (O – E)				
$(O - E)^2$				
$(O - E)^2 / E$				
$\chi^2 = \Sigma [(O - E)^2 / E]$				

Table 4.2

	Probability, p				
Degrees of freedom	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Table 4.3

- (iii) With reference to the table of probabilities, **Table 4.3**, explain how the value for the chi-squared test supports the hypothesis that these are two pairs of segregating alleles at two loci. [2]

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[Total: 9 marks]

Question 5

With the completion of the Human Genome Project, it was estimated that the human genome was likely to contain only 30,000 genes. Counting the number of protein-encoding genes in the human genome is a difficult process, as the “start” and “stop” boundaries of transcribed genes are only imperfectly understood.

A young scientist has decided to take a different approach of capturing mRNAs from the cytoplasm, before converting them to DNA to estimate the number of genes present. Using this approach, he reports that there are over 65,000 discrete mRNA-transcribing sequences in the genome, hence there are more genes than estimated by the Human Genome Project.

- (a) Describe briefly how post-transcriptional modification would allow all types of mRNA to be captured using a single method, and how mRNA can be subsequently converted to DNA. [2]

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- (b) State why counting the number of unique mRNA sequences will result in an over-estimate of the number of actual genes in an organism. [1]

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- (c) Within the same human cell, the level of gene expression can differ greatly between genes. Briefly describe what could account for the different levels of mRNA produced for different genes. [2]

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(d) The eukaryote genome differs from the prokaryote genome in that some genes may be present in multiple copies, a result of gene amplification.

(i) Describe briefly one process by which gene amplification can occur. [2]

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(ii) In nature, some organisms are able to amplify certain genes at some stage in their development. Describe the significance of gene amplification in development. [2]

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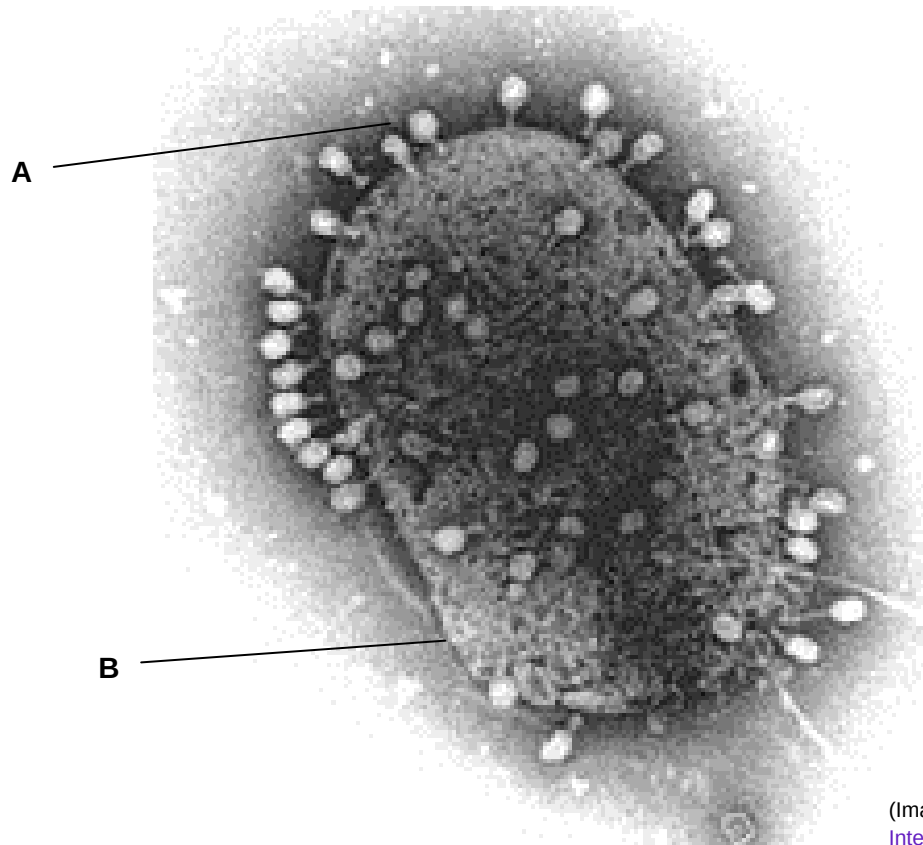
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[Total: 9 marks]

Question 6

Fig. 6.1 shows a scanning electron micrograph of *Escherichia coli* under mass attack by numerous phage-T4 virions.



(Image courtesy of [Cornell Integrated Microscopy Center](#))

Fig. 6.1

(a)(i) Suggest how **A** recognizes the correct host cell. [1]

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(ii) With reference to **Fig. 6.1**, describe a process whereby bacterial DNA can be transferred from one cell to another. [2]

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- (iii) Give **two** differences between **2 other processes** (besides that one named in (aii)) that result in genetic variation in bacteria. [3]

Process 1: Process 2:

Differences:

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Fig. 6.2 shows the Jacob and Monod model of gene expression in *Escherichia coli*, the lac operon model.

(Source: <http://www.ccrnp.ncifcrf.gov/~toms/genetic.control/www.uic.edu/lect15.htm>)

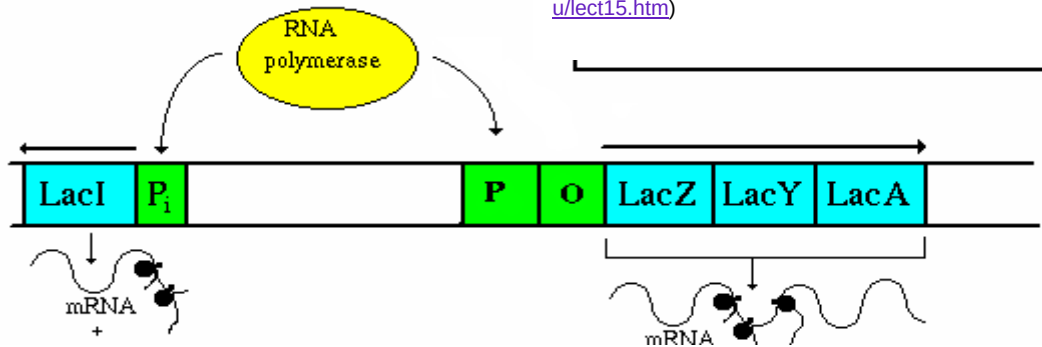


Fig. 6.2

- (b) Explain the following terms:

- (i) RNA polymerase. [1]

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- (ii) Operator O. [1]

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- (c) With reference to **Fig. 6.2**, distinguish clearly between a regulatory gene and a structural gene. [2]

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- (d) Are the enzymes produced by the lac operon inducible or repressible? Explain your answer. [2]

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[Total: 12 marks]

Question 7

Fig. 7.1 shows two motor neurons, **A** and **B**. It also shows the synapses of neuron **B** with two other neurons, **P** and **Q**.

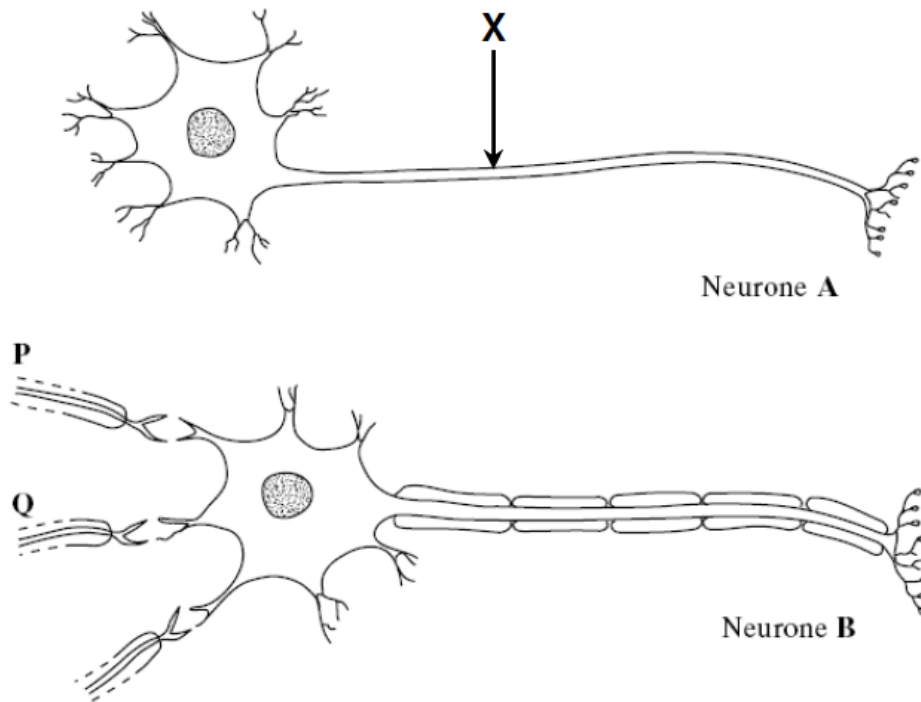


Fig. 7.1

- (a)(i) If an action potential is initiated at point **X** on neuron **A**, describe how this action potential is propagated along the neuron. [3]

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- (ii) Suggest why transmission of impulses along neuron **B** uses less energy than transmission along neuron **A**. [2]

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GABA is a neurotransmitter which inhibits the production of action potentials.

Fig. 7.2 and the **graph 7.3** show how the release of GABA from a presynaptic membrane affects the membrane potential of a postsynaptic membrane.

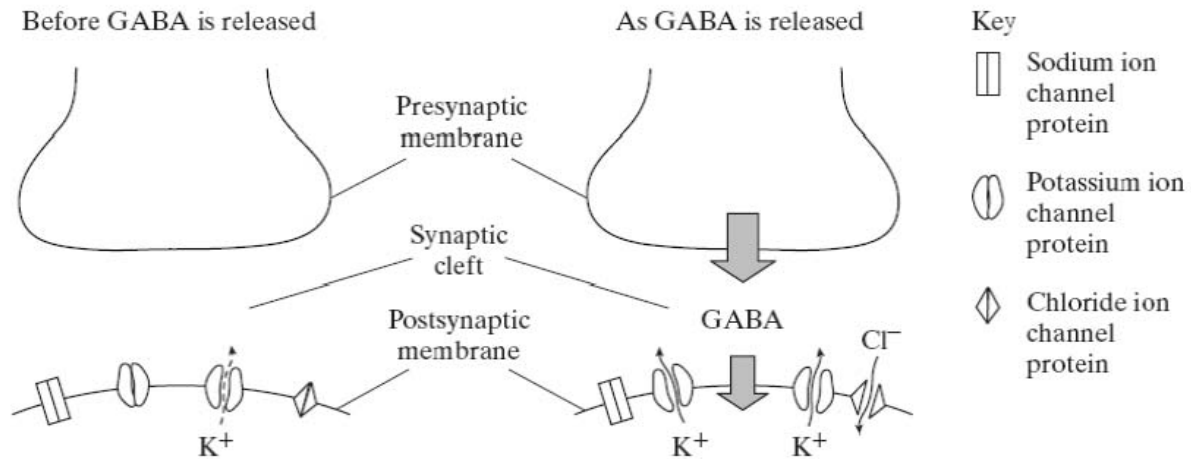
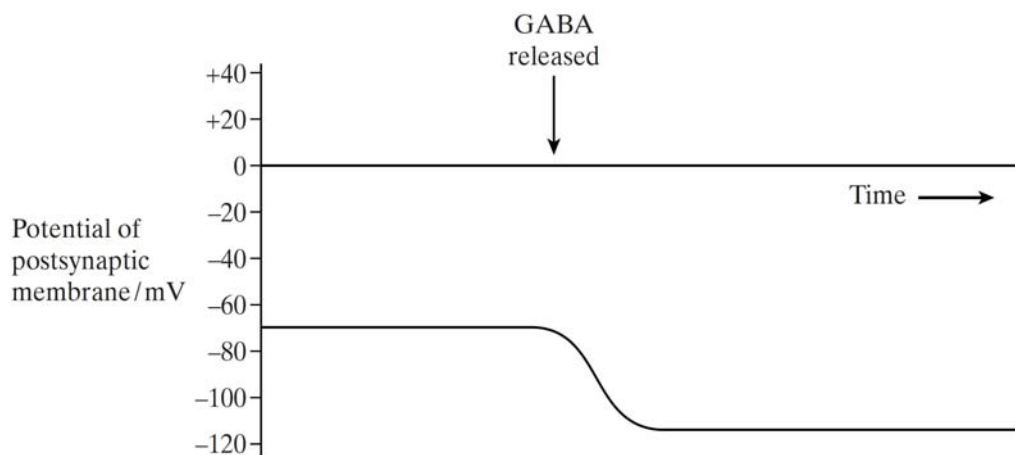


Fig. 7.2



Graph 7.3

- (b) When the postsynaptic membrane is stimulated by acetylcholine, an action potential is less likely to occur if GABA is released at the same time. Explain why. [2]

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Glucagon, like GABA and acetylcholine, is an essential first messenger molecule in the signal transduction needed in the regulation of blood glucose level in our body. Information from glucagon is transduced as soon as it binds to cellular surface receptors.

- (c)(i) Explain the roles of adenylate cyclase, in signal transduction, that leads to the breakdown of glycogen in liver cells. [2]

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- (ii) Suggest the importance of protein kinases and phosphatases in phosphorylation cascades. [2]

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[Total: 11 marks]

Question 8

The Galapagos are a very isolated group of islands. Species of birds called finches live on these islands. Scientists think that the different species have evolved from a population of one species which managed to reach the islands thousands of years ago from South American mainland. Scientists used base sequences in the DNA of the finches to work out the evolutionary relationship between the species. **Fig. 8.1** shows the evolutionary relationship.

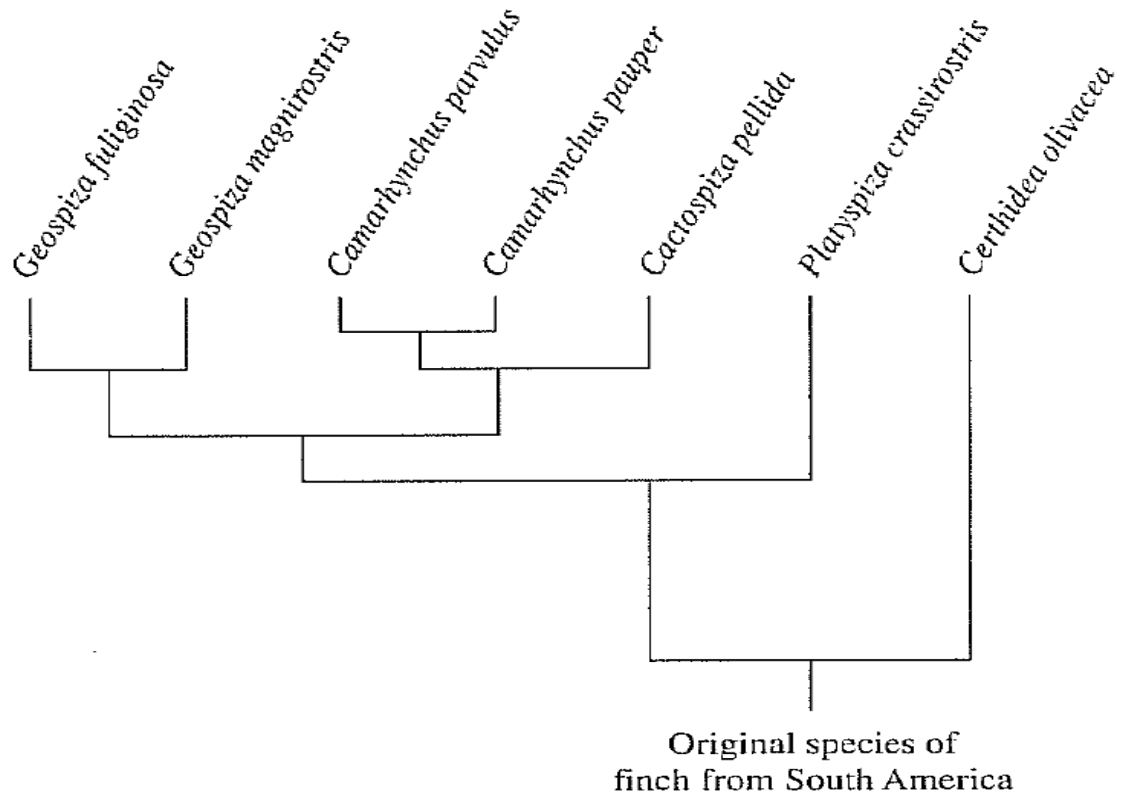


Fig. 8.1. Evolutionary Relationship between the Finches

- (a) Using the information in the diagram, state the present-day species of finch that evolved first on the Galapagos islands? Explain your answer. [2]

Finch species:

Explanation:

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- (b) Explain why evaluating the DNA base sequences can work out the evolutionary relationship between the species of finches. [2]

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- (c) The beak of each species is adapted for feeding on a particular type of food. *Geospiza fuliginosa* evolved as an eater of small seeds. *Geospiza magnirostris* evolved on a different island as an eater of large seeds.

- (i) Suggest how these different species of finch might have evolved from one species. [4]

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- (ii) Describe how biogeography may provide support for the evolution of the different species of finch. [2]

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[Total: 10 marks]

Section B (20 marks)

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Question 9

- (a) Compare and contrast Neo-Darwinism with Neutral Theory. [4]
- (b) Describe how the membrane affects the entry and exit of substances in and out of a cell.[6]
- (c) Define control elements and explain how they influence transcription. [10]

Question 10

- (a) Describe the mechanism of synaptic transmission and contrast this with the transmission of a nerve impulse. [6]
- (b) With reference to the three different stages of cell signaling, describe the series of reactions that take place after a glucagon hormone binds to its target cell, causing the overall increase of blood sugar level. [8]
- (c) Explain the small yield of ATP from anaerobic respiration. [6]

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